Measures of Association

Scatter diagrams, covariance, Pearson's and Spearman's correlation, Kendall's concordance and correlation matrices.

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STAT 3202 · #KAUMeLON · Official revision sheet

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1 Relationships & the scatter diagram

Linear
Can be represented by a straight line.

Monotonic
One variable consistently rises (or falls) as the other rises, not necessarily at a constant rate.

A **scatter diagram** plots paired observations (x_i, y_i): always look at it before computing a coefficient.

2 Types of correlation

Positive	Move in the same direction
Negative	Move in opposite directions
Simple	Two variables
Multiple	One variable with a set of others
Partial	Two variables, controlling for others
Linear / non-linear	Straight-line / curved pattern

3 Karl Pearson's correlation coefficient r

Covariance

$$cov(x,y) = \frac{1}{n}\sum(x_i - \bar{x})(y_i - \bar{y})$$

Correlation

$$r = \frac{cov(x,y)}{sd(x)\, sd(y)}$$

Deviation form

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

Measures **linear** association. Covariance depends on units; r is unit-free. r is sensitive to outliers.

4 Reading r

+1	Perfect positive
near +1	Strong positive
0	No **linear** correlation
near −1	Strong negative
−1	Perfect negative

$-1 \le r \le +1$. Perfect: all points on a line.

5 Spearman's rank correlation ρ

Computed from **ranks**; measures **monotonic** association. Good for ordinal data and curved but monotonic patterns. Range −1 to +1.

No tied ranks

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$$

d_i = difference between the ranks of the ith pair. Ties get their **average rank** and correction terms $T = \frac{1}{12}\sum m(m^2-1)$.

6 Kendall's coefficient of concordance W

Measures **agreement among several judges' rankings** (e.g. k judges ranking n varieties).

Concordance

$$W = \frac{12\left[\sum R_i^2 - \frac{(\sum R_i)^2}{n}\right]}{k^2 n(n^2-1)}$$

R_i = sum of ranks for object i; k = number of judges.

7 Correlation matrix & correlogram

Correlation matrix
Table of pairwise correlations among several variables. Diagonal elements = **1**; symmetric about the diagonal.

Correlogram
A graphical display of the correlation matrix (colour or size shows strength and sign).

	Pearson r	Spearman ρ
Uses	Actual values	Ranks
Detects	Linear association	Monotonic association
Outliers	Sensitive	Less sensitive

! Common mistakes

- × Correlation is **not causation**.
- × $r = 0$ rules out only **linear** association; a curve can still exist.
- × Skipping the scatter plot.
- × Forgetting average ranks for ties.

? Fill in the blanks

1. If cov(x, y) = 12, sd(x) = 3 and sd(y) = 5, then r = _____. [0.8]
2. If all points lie exactly on a falling straight line, r = _____. [−1]
3. For n = 10 pairs with Σd² = 33, Spearman's ρ = _____. [0.8]
4. Two judges who rank five varieties in exactly the same order give ρ = _____. [1]
5. Changing yield from kg to quintals changes r by _____. [nothing (r is unit-free)]
6. r = 0 rules out a straight-line relationship but not a _____ one. [curved (non-linear)]

Source: *Textbook of Agricultural Statistics*, Ch. 8 · Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul · KAU Me-LON

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